SAKSHAM UBOWEJA

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EDUCATION

University of California at Berkeley Berkeley, California
BA in Applied Mathematics & Computer Science Graduation: 2029
Relevant Coursework: Data Structures & Algorithms, Linear Algebra, Multivariable Calculus, Microeconomics, Macroeconomics

EXPERIENCE

NeurIPS Jul 2025 – Sep 2025
Programmer & Researcher
- Co-authored the paper introducing the first scalable methodology of improving CoT faithfulness, implementing a PyTorch DPO training loop over paired chain-of-thought traces with robust data loaders, filtering, and an evaluation harness.
- Designed faithfulness and consistency metrics, integrated experiment tracking and automated analysis notebooks, and demonstrated notable accuracy and faithfulness gains over standard CoT prompting; presenting results at the NeurIPS 2025 FoRLM workshop.

University of Oxford Aug 2023 – Aug 2025
Data Analyst & Assistant Researcher (mentored by Dr. Alexander Mushtukov)
- Modeled black-hole accretion-disk broadband variability with NumPy/Matplotlib, running 150+ stochastic simulations to compare synthetic power spectra with observational data.
- Applied 3D linear algebra and Green's functions to photon-trajectory simulations and added unit-style numerical stability checks, increasing the robustness of the analysis pipeline.

University of Cambridge Nov 2023 – Sep 2024
Lead Researcher (mentored by Dr. Damián Pitalúa-García)
- Led GO/ZnO quantum-dot synthesis runs and scripted comparisons of charge-separation characteristics; produced publication-quality figures directly from analysis notebooks.
- Evaluated QD-doped Li-polymer cells that showed a 20–30% battery-life gain versus baseline cells, informing follow-on experiments and paper drafts.

Project Streamline May 2023 – Nov 2024
Co-Founder & CEO
- Founded a privacy-focused biometric authentication startup using zero-knowledge proofs so organizations can verify attendance and access without storing raw biometric data, targeting an identity and access management market of approximately $20B.
- Represented Streamline at GITEX and Intersec (events with 180,000+ and 50,000+ attendees, respectively), pitching to security and technology executives; converted follow-ups from these conferences into a ∼60% increase in the number of enterprise and technology contacts relative to pre-event outreach, including partnership discussions with cloud providers and biometric hardware vendors.

ACTIVITIES & LEADERSHIP

Cygfix Apr 2023 – Apr 2025
Founder & CEO
- Founded a student-run circular-economy initiative in the UAE that collects and repurposes 50,000+ cigarette butts per month into new materials; recognized by *Khaleej Times* for its sustainability impact.
- Designed and rolled out a passive-collection model at 20+ high-traffic sites (cafés, promenades, and campuses) and led outreach campaigns reaching 4,000+ residents to shift norms around cigarette littering.

Modern Grandmasters Chess Club Apr 2023 – Dec 2024
President & Chief Arbiter
- Expanded active membership by ∼40% over two years by organizing 10+ tournaments and training camps and mentoring club players, some of whom progressed to national-level representation.

iCAN Global (Design for Change) Jul 2023 – Nov 2024
Lead Partner & International Manager
- Co-led UAE national rounds (60+ participants) and coordinated the 2024 global innovation program for 500+ students from 40+ countries, managing communication, schedules, and judging logistics across time zones.

AWARDS

UAE Golden Visa Recipient for Academic Excellence | IB World Topper in Math, Physics, Chemistry & Economics | National Olympiad: Physics (Rank-5), Chemistry (Rank-6) | 1st Runner-Up, JGI International Research Competition | National Academic Merit Scholar

SKILLS

Programming Languages: Python, Java
Python Libraries: PyTorch, NumPy, Jupyter Notebook
Spoken Languages: English, Hindi, Punjabi, Arabic